Voice Emotion Recognition using Audio Signal Processing and Machine Learning

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Introduction

- Emotional awareness and mental wellness are increasingly important in today's digital world.
- A Voice Diary System with emotion recognition provides an innovative way to record personal experiences while automatically detecting underlying emotions.
- Voice diaries capture tone, pitch, intensity, and pauses, offering deeper emotional insight than traditional written journals.
- Digital Signal Processing (DSP) and Machine Learning (ML) techniques are applied to classify emotions such as happy, sad, angry, fearful, surprised, and neutral from speech signals.
- This technology can be applied to call centers, healthcare, smart assistants, and security systems to make interactions more emotionally intelligent and responsive.

Problem Statement

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Traditional written and digital journals lack the ability to capture vocal nuances like tone, pitch, and intensity, creating a need for a Voice Diary System that uses Audio Signal Processing and Machine Learning to objectively detect emotions and provide deeper emotional insights for mental wellness and empathetic AI interactions

Objectives

Objective

- To design and develop a voice diary application that allows users to record and store daily audio journal entries.
- To implement a Speech Emotion Recognition (SER) pipeline that analyzes speech signals and predicts emotional states such as happy, sad, angry, fearful, and neutral.
- To generate emotion analytics such as mood distribution, weekly/monthly emotion trends, and emotional pattern tracking over time.
- To build a user-friendly dashboard that helps users visualize and understand their emotional well-being through their voice diary recordings.
- Implement a Secure Backend for Data Persistence and Model Management

Literature Review

No.	Title	Methodology	Gaps Identified	Ideas Adopted
1.	Analysis of Speech Emotion Recognition and Detection using Deep Learning, 2022 [1]	Implements a standard SER workflow using preprocessing and MFCC extraction. Uses an LSTM model to classify emotions from speech features. Uses spectrogram and waveform analysis to explain emotional cues. Evaluates emotion detection using categorical emotion classes.	Provides limited novelty beyond standard SER implementations. Does not address accent variation and cross-speaker generalization. Does not focus on emotion analytics or diary-based user applications.	MFCC + LSTM is a reliable baseline for SER implementation. Spectrogram analysis is useful for explaining emotional patterns in speech. SER systems can be structured as modular pipelines for easier development.

No.	Title	Methodology	Gaps Identified	Ideas Adopted
2.	Voice Recognition and Voice Comparison using Machine Learning Techniques: A Survey, 2020 [2]	Surveys traditional voice recognition and comparison approaches. Reviews common speech feature extraction techniques like MFCC and LPC. Summarizes ML classifiers such as SVM, k-NN, and GMM. Discusses practical challenges like noise, channel variation, and dataset limitations.	Mainly focuses on speaker recognition, not speech emotion recognition. Provides limited SER-specific implementation details. Does not address emotion trend analysis or diary-style applications.	Classical ML models can serve as efficient baselines for comparison. Feature quality plays a major role in system accuracy. Noise and recording conditions must be addressed in real-world voice systems.

No.	Title	Methodology	Gaps Identified	Ideas Adopted
3.	Speech Emotion Recognition Using Deep Learning, 2024 [3]	Builds an SER pipeline using speech preprocessing and feature extraction. Uses MFCC features to represent emotional cues in speech. Trains an LSTM-based deep learning model for emotion classification. Uses supervised learning to predict categorical emotions from speech.	Limited emphasis on noise robustness and real-world recording conditions. Does not focus on real-time constraints or deployment considerations. Does not connect SER predictions to user analytics or mood tracking.	LSTM models are useful because emotions evolve over time in speech. MFCC is a widely used and practical representation for SER tasks. Deep learning improves emotion recognition compared to simpler approaches.

No.	Title	Methodology	Gaps Identified	Ideas Adopted
4.	Voice Feature Extraction Method Analysis for Speaker Recognition with Degraded Human Voice, 2023 [4]	Extracts MFCC features from speech signals. Uses Gaussian Mixture Model (GMM) for classification. Tests system performance under degraded/noisy speech conditions. Analyzes impact of sampling rate and noise levels on recognition.	Focuses on speaker recognition rather than emotion recognition. Does not include deep learning architectures like CNN/LSTM. Does not address emotion ambiguity or multi-class emotion analytics.	MFCC can be used as a strong baseline feature for speech-based tasks. Noise directly affects feature stability and classification performance. Real-world systems must consider degraded audio quality.

No.	Title	Methodology	Gaps Identified	Ideas Adopted
5.	Real-time Speech Emotion Recognition using Deep Learning and Data Augmentation, 2024 [5]	Proposes a real-time SER system using deep learning approaches. Extracts features such as MFCC, ZCR, RMS, chroma, and Mel spectrograms. Trains models including MLP, CNN, and hybrid CNN+BiLSTM architectures. Uses data augmentation methods like noise addition and spectrogram shifting.	Focuses mainly on recognition accuracy rather than user analytics. Model complexity may increase computational cost for lightweight devices. Does not explore personalization for different users' speaking styles.	Data augmentation improves robustness and reduces overfitting in SER. CNN+BiLSTM is effective for learning both spectral and temporal emotion cues. Real-time SER design is relevant for practical applications like voice diaries.

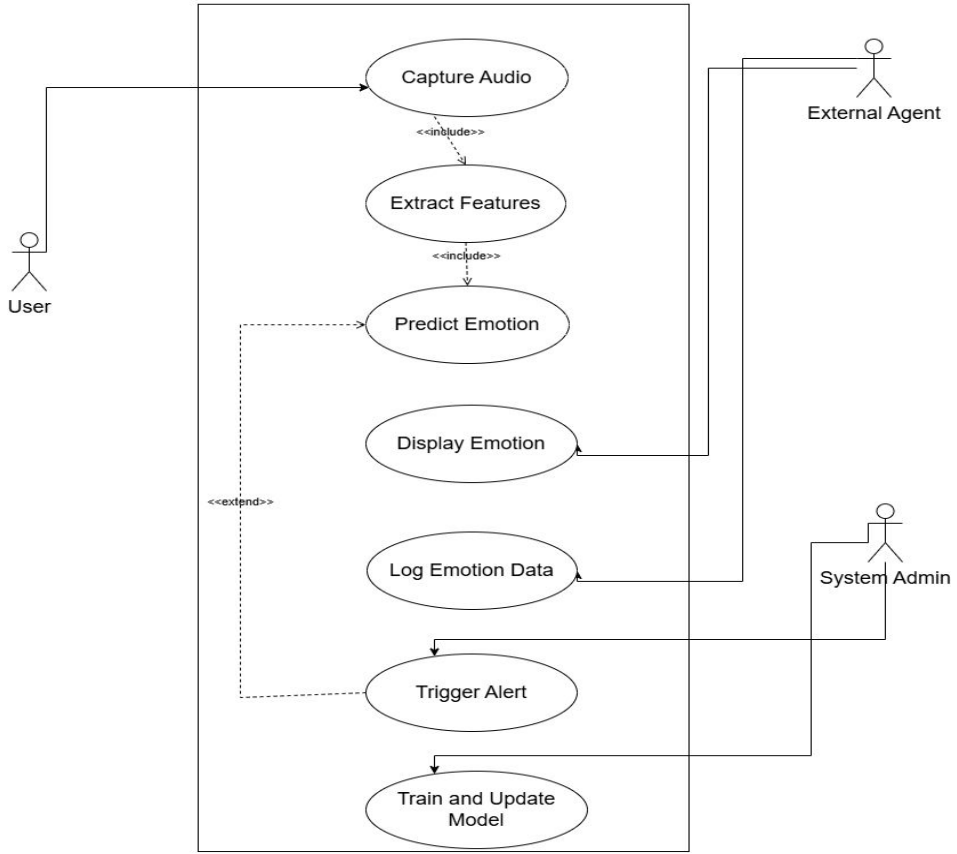
No.	Title	Methodology	Gaps Identified	Ideas Adopted
6.	Curriculum Learning for Speech Emotion Recognition From Crowdsourced Labels, 2019 [6]	Introduces curriculum learning for training deep SER models. Defines training difficulty using disagreement among human annotators. Trains models from easy samples to difficult samples for better learning. Demonstrates the approach for regression, binary, and multi-class SER tasks.	Requires multi-rater emotion annotations, which may not always be available. Mainly improves training strategy rather than application design. Does not focus on real-time SER deployment or user-facing dashboards.	Emotion ambiguity can be treated as a measurable difficulty factor. Training on clearer samples first can improve model generalization. Annotator agreement can be used as a reliability indicator in SER datasets.

No.	Title	Methodology	Gaps Identified	Ideas Adopted
7.	A Comprehensive Review of Speech Emotion Recognition Systems, 2021 [7]	Surveys complete SER systems from preprocessing to classification. Reviews feature extraction methods including MFCC and spectrogram-based inputs. Compares classical ML approaches with deep learning-based SER models. Summarizes major challenges such as noise, speaker variability, and limited data.	Survey paper does not provide a single reproducible system design. Does not focus on voice diary applications or emotion analytics. Provides limited implementation-level details for model building.	Using both ML and deep learning provides meaningful baseline comparisons. Real-world SER must handle noise, accent variation, and recording differences. SER is best designed as an end-to-end pipeline with clear modules.

No.	Title	Methodology	Gaps Identified	Ideas Adopted
8.	Head Fusion: Improving the Accuracy and Robustness of Speech Emotion Recognition, 2021. [8]	Proposes a CNN-based SER model enhanced with multi-head attention. Introduces “Head Fusion” to improve emotion classification accuracy. Tests robustness by injecting many types of real environmental noises. Studies how noise intensity and noise type affect SER performance.	Attention-based models may be heavy for low-resource deployment. Does not provide application-level integration like diary analytics. Increased complexity makes reproduction harder compared to simpler baselines.	Attention mechanisms improve SER by focusing on emotion-relevant speech regions. Noise robustness must be tested explicitly for real-world deployment. Training with carefully selected noise can improve generalization.

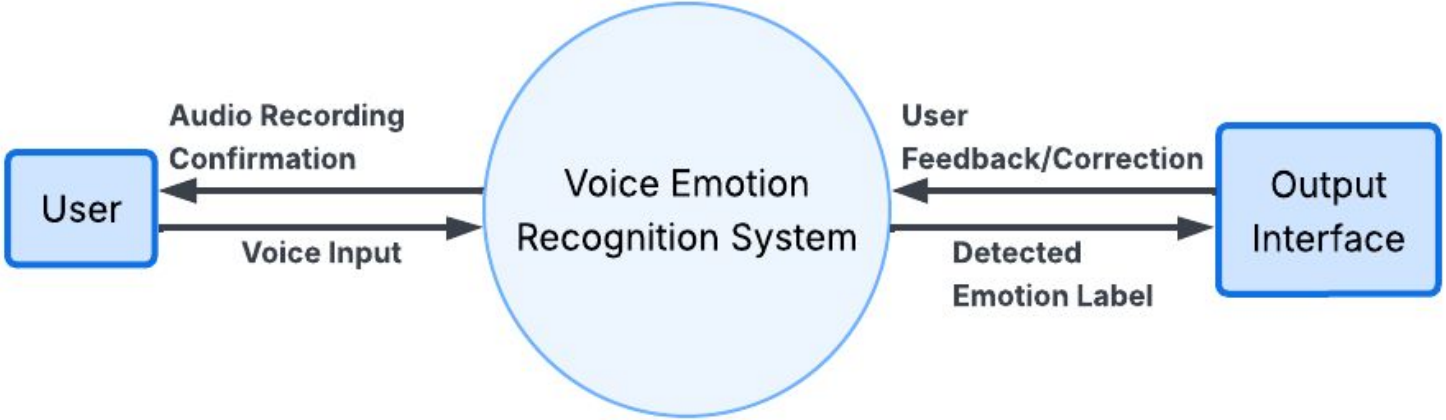
Use Case Diagram

Voice Emotion Recognition System

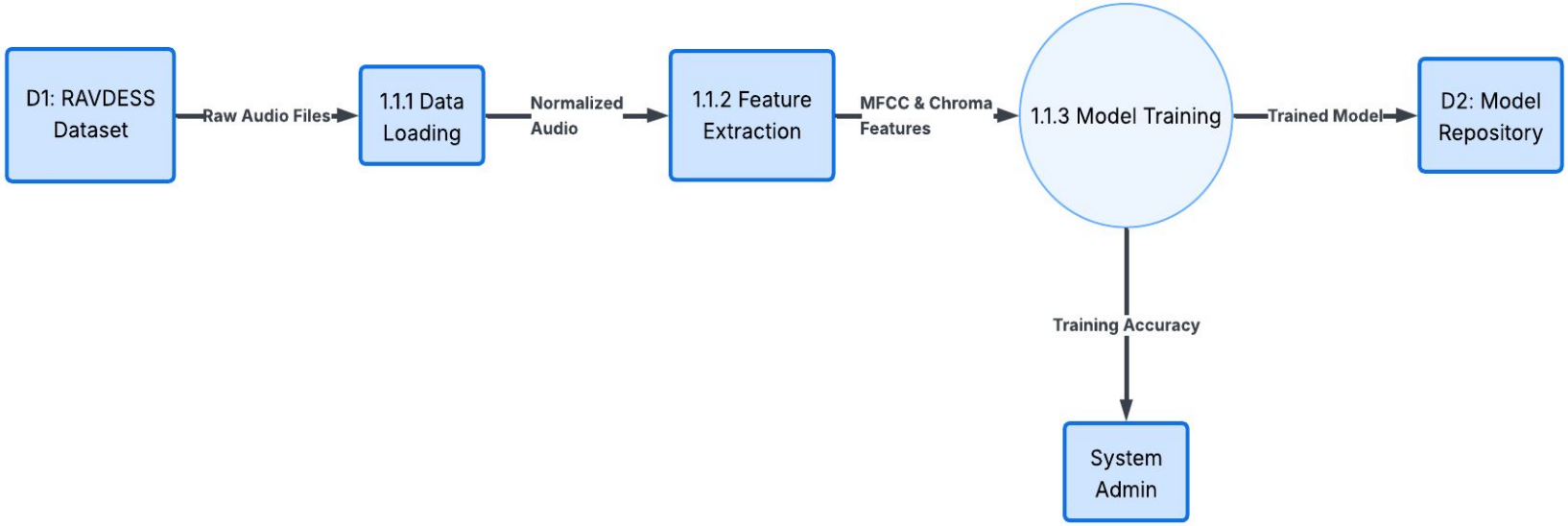


Data Flow Diagram

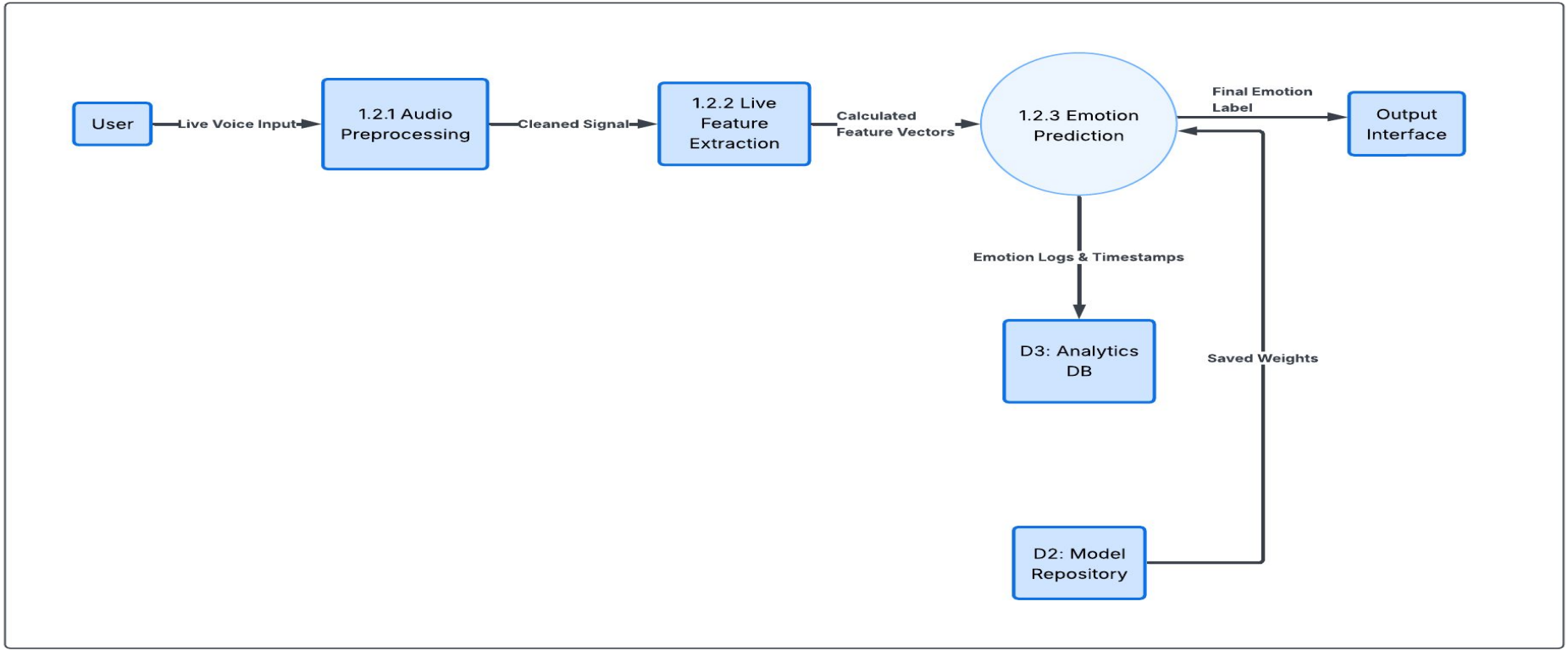
Voice Emotion Recognition DFD Level 0



Model Training DFD Level 1.1



Inference Phase DFD Level 1.2



Proposed Methodology

Proposed Methodology

1. Voice Diary Application Development

- **Audio Capture**: Interface with a microphone to record human speech as raw audio signals.
- **Storage Workflow**: Develop a system to save and organize these voice recordings as daily diary entries

2. Speech Emotion Recognition (SER) Pipeline

- **Preprocessing**: Use Digital Signal Processing (DSP) to clean signals and remove background noise.

- **Feature Extraction**: Utilize the Librosa library to extract numerical features such as MFCC, Chroma, and Mel Spectrograms.

- **Classification**: Apply Machine Learning or Deep Learning models (like CNN or LSTM) to predict emotions such as happy, sad, or angry.

Proposed Methodology

3. **Emotion Analytics Generation**

- **Data Aggregation**: Collect detected emotion labels and timestamps from stored entries.
- **Trend Analysis**: Process the logged data to identify mood distributions and weekly or monthly emotional patterns.

4. **User-Friendly Dashboard**

- **Visualization**: Design an interface to display detected emotions and graphical insights to the user.
- **Well-being Insights**: Present emotional trends and history to help users understand their mental wellness.

Proposed Methodology

5. Secure Backend and Model Management

- **Data Persistence**: Implement a database to securely store audio files, emotion logs, and diary metadata.
- **Model Repository**: Create a storage system to save trained model weights (e.g., .h5 or .pkl files) for real-time deployment

Conclusion

The proposed Voice Diary with Emotion Recognition system combines audio journaling with Artificial Intelligence to create an emotionally aware digital solution. By using DSP techniques and ML models, it can analyze speech patterns and accurately classify human emotions.

This system enhances self-reflection, supports mental wellness, and improves human-AI interaction. With future enhancements such as multilingual support, real-time inference, and mobile or edge deployment, it has strong potential for practical applications across various domains.

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Thank You!